Test Plan

VWO Login Dashboard (app.vwo.com/login)

Prepared for	SmartBear
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1. Objective

The goal of this test plan is to ensure the quality, functionality, security, accessibility and performance of the VWO (Visual Website Optimizer) login dashboard at https://app.vwo.com/login, the entry point to VWO's experimentation, personalization and analytics tools.

The plan verifies the PRD's business objectives: secure access to the platform, minimal login friction, support for enterprise security requirements and compliance, and seamless onboarding of new users.

Success Metrics to Validate (from PRD)

Category	Target
Performance	Login success rate 95%+; login page load under 2 seconds; 90%+ user satisfaction
Security	Zero successful brute force attacks or unauthorized access; 100% compliance with security audit requirements; no session hijacking
Business	Improved retention; increased trial-to-paid conversion; 20% fewer login-related support tickets
Availability	99.9% uptime

2. Scope

In Scope

1. **Functional Testing** – email/password login, session management (configurable timeout), Remember Me, optional 2FA, enterprise SSO, social login, forgot password / recovery, registration (free trial) link, transition to the VWO core dashboard.
2. **Data Validation Testing** – validation on blur, email format verification, password strength indicators, password requirements.
3. **Error Handling Testing** – clear, actionable messages for failed authentication; error recovery flow; success confirmation.
4. **UI / Usability Testing** – responsive design, auto-focus, clickable labels, loading states, branding consistency, product announcement banner, Light and Dark Mode.
5. **Accessibility Testing** – screen reader support (ARIA labels), keyboard navigation, high contrast mode, WCAG 2.1 AA.
6. **Security Testing** – encryption of authentication data, HTTPS/TLS enforcement, hashed password storage, secure session tokens, rate limiting against brute force, OWASP authentication guidelines.

7. **Compliance Testing** – GDPR and CCPA handling of user data, enterprise security policies and audit trails.
8. **Performance and Load Testing** – page load within 2 seconds, asset compression and minification, CDN usage, thousands of simultaneous login attempts.
9. **Availability / Scalability Testing** – 99.9% uptime, multi-region performance, auto-scaling under traffic spikes.
10. **Integration Testing** – VWO core platform, login success/failure analytics tracking, customer support integration, SAML/OAuth SSO, Google/Microsoft social login, marketing/onboarding tools.
11. **Compatibility Testing** – browsers, operating systems and devices (see Test Environments).
12. **Regression Testing** – after fixes and after each development phase.
13. **Ad Hoc / Exploratory Testing.**

Out of Scope (PRD "Future Enhancements")

- Biometric authentication (fingerprint, facial recognition)
- Adaptive (risk-based) authentication
- Progressive Web App experience
- A/B testing, behavior analytics and personalization of the login experience

3. Inclusions

Authentication System

- Login with valid and invalid email/password combinations.
- Session creation, timeout and token security; Remember Me persistence.
- Optional 2FA setup and verification.
- Enterprise SSO (SAML, OAuth) and social login (Google, Microsoft).

User Input Validation

- Real-time validation on blur; email format and mobile keyboard type.
- Password strength indicator and complexity requirements.
- Empty, malformed, boundary-length and special-character inputs.

Password Management

- Forgot password flow, secure token generation, email-based reset, token expiry/reuse.

User Experience and Accessibility

- Responsive layout and touch-friendly controls; auto-focus; clickable labels; loading state.
- Screen reader, keyboard-only navigation, high contrast, Light/Dark Mode.

Security and Compliance

- HTTPS enforcement; encryption in transit; injection (SQL, XSS) and brute-force attempts; rate limiting; session hijacking.
- GDPR/CCPA data handling; audit trail support.

Performance and Scalability

- Page load time; concurrent login load; uptime; CDN and multi-region behavior.

Integration

- Redirect to main dashboard and context preservation of previous sessions.

- Login success/failure tracking; support system link; registration/free trial path.

User Journeys

- New user: discovery → registration CTA → onboarding.
- Returning user: quick access → dashboard transition → recent activity.
- Error recovery: error identification → recovery options → success confirmation.

4. Test Environments

Name	Env URL
Production login page (under test)	https://app.vwo.com/login
QA	*To be confirmed*
Pre Prod	*To be confirmed*

Platforms and Browsers

- Windows 10 – Chrome, Firefox and Edge
- Mac OS – Safari
- Android Mobile OS – Chrome
- iPhone Mobile OS – Safari

Other Environment Details

- Device types: desktop, laptop, tablet and smartphone, with varied screen sizes.
- Network: Wi-Fi, cellular and wired, including slow connections for the 2-second load target.
- Access: test accounts (standard, 2FA-enabled, SSO) and roles for testers, developers and stakeholders – *to be provided*.

5. Defect Reporting Procedure

- **Defect criteria:** deviation from the PRD, user experience issue, or technical error.
- **Reporting steps:** log in JIRA using the defect template with reproduction steps, expected vs. actual result, environment/browser, and screenshots or logs.
- **Triage:** assign severity and priority and route to the right owner for investigation and resolution.
- **Communication:** defects found each day are sent to development management in an end-of-day status email.
- **Metrics:** defects found, time to resolve, percentage fixed.

Defect Area	POC
Frontend	*TBD*
Backend	*TBD*
Dev Ops	*TBD*

Tool: JIRA

6. Test Strategy

Step 1 – Test Design

- Create test scenarios and test cases for each feature in scope, and prioritize them.
- Design techniques: Equivalence Class Partition, Boundary Value Analysis, Decision Table Testing, State Transition Testing, Use Case Testing.
- Experience-based: Error Guessing and Exploratory Testing.

Step 2 – Test Execution Process

- Smoke testing first on each build to check that the important login functionality works.
- If smoke testing fails, reject the build and wait for a stable one.
- On a stable build, perform in-depth testing with the created test cases.
- Multiple testers work on multiple supported environments simultaneously.
- Report bugs in the bug tracking tool and send the end-of-day status email.
- Types of testing: Smoke and Sanity, Regression and Retesting, Usability, Functionality and UI; repeat test cycles until quality is achieved.

Step 3 – Best Practices

- Context Driven Testing.
- Shift Left Testing – start from the early development stages.
- Exploratory Testing alongside scripted cases.
- End to End Flow Testing – e.g., register → login → dashboard; failed login → recovery → successful login.

Alignment with PRD Development Phases

Phase	Test Focus
Phase 1: Core Authentication	Login form, basic validation and error handling, password reset
Phase 2: Enhanced UX	Responsive design, accessibility, advanced validation and feedback
Phase 3: Enterprise Features	SSO, advanced security, analytics and monitoring

7. Test Schedule

The application will be tested over **2 Sprints**.

Task	Dates
Creating Test Plan	*TBD*
Test Case Creation	*TBD*
Test Case Execution	*TBD*
Summary Reports Submission	*TBD*

8. Test Deliverables

- Test Plan

- Test Scenarios
- Test Cases
- Defect Reports
- Test Case Execution Reports
- Test Summary Report

9. Entry and Exit Criteria

Requirement Analysis

Entry: The testing team receives the requirements document (this PRD).
Exit: Requirements are explored and understood by the testing team; doubts are cleared.

10. Test Execution

Entry: Test scenarios and test cases are signed off by the client; application is ready for testing.
Exit: Test case reports and defect reports are ready.

11. Test Closure

Entry: Test case reports and defect reports are ready.
Exit: Test summary report is delivered.

12. Tools

- JIRA bug tracking tool
- Mind map tool
- Snipping screenshot tool
- Word and Excel documents

13. Risks and Mitigations

Risk	Mitigation
Non-availability of a resource	Backup resource planning
Build URL is not working	Resources will work on other tasks
Less time for testing	Ramp up resources dynamically based on client needs
Security risks (PRD)	Regular security audits, penetration testing, real-time monitoring and patching
Performance risks (PRD)	Load testing under varied conditions, real-time performance monitoring, auto-scaling

14. Approvals

The team will send the following documents for client approval. Testing continues to the next step only once approvals are done.

- Test Plan
- Test Scenarios
- Test Cases
- Reports

Name	Role	Signature	Date